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# Macroeconomic uncertainty and expected shortfall (and value at risk): a new dynamic semiparametric model

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***Macroeconomic uncertainty variables have significant effects on long term components of stock price volatility and can be used to improve VaR and ES prediction for a given stock***

## 1. Introduction

VaR is a standard tool for measuring market risk. Basel III places new emphasis on ES as a partial substitute for and complement to VaR (Basel Committee 2010). However, the applications of ES are much fewer than those of VaR, despite the former being widely accepted as a 'coherent' measure of risk (Artzner *et al.* 1999). One of the important deterrents to using this measure is that it is not 'elicitable' (Patton *et al.* 2019). In other words, there are no loss functions such that ES is the solution to minimizing the expected loss (Gneiting 2011). This causes difficulties in estimating ES on dynamic models (Rockafellar *et al.* 2014). Recently, Fissler and Ziegel (2016) show that ES and VaR are jointly elicitable by proposing a new family of loss functions, which provides a foundation for modeling ES and VaR. Based on

this framework, Taylor (2019) uses the Laplace distribution to estimate dynamic models for VaR and ES jointly. Patton *et al.* (2019) (PZC hereafter) propose another general class of dynamic semiparametric models for ES and VaR, and also give an asymptotic theory for these types of models.

In this paper, we extend the class of the models of Patton *et al.* (2019). We make two contributions to the literature. As a theoretical contribution, we account for the effects of macroeconomic variables on ES and VaR in the proposed model. The PZC model uses the generalized autoregressive score (GAS) of Creal *et al.* (2013) and Harvey (2013) or the generalized autoregressive heteroskedasticity (GARCH) of Bollerslev (1986) to capture the volatility dynamics of financial asset returns. Using the PZC model, Lazar and Zhang (2019) investigate the model risk of ES. Meng and Taylor (2020) estimate VaR and ES using intraday low and range data. Our GAS model allows for mixed-frequency data using the GARCH-MIDAS

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specification of Engle *et al.* (2013) which can capture the impacts of low-frequency macroeconomic variables on high-frequency volatility. The GARCH-MIDAS model decomposes the volatility process into two components: a short-term volatility component, captured by GARCH, and a long-term component, captured by a mixed-frequency data sampling (MIDAS) regression. Following the GAS approach, Massacci (2017) models the tails of the conditional distribution of financial returns using results from extreme value theory, and link the risk dynamics to the macroeconomy. Our model differs from Massacci (2017) in the way that it accounts for the role of macroeconomic indicators directly by incorporating them as the driving factor of volatility components.

We provide the asymptotic theory of a new class of dynamic semiparametric models. Most of the conditions under which the estimators are asymptotically normal and consistent can be found in the literature on the asymptotic properties of GARCH-type models (e.g. Lumsdaine 1996, Carrasco and Chen 2002, Patton *et al.* 2019). We also provide estimates of the asymptotic covariance. The results of a Monte Carlo simulation study indicate that estimates based on FZ minimization are much less sensitive to a change in the data generating process (DGP) in comparison with the maximum likelihood estimates. The proofs of the asymptotic properties of the estimates are based on the PZC paper.

Our second contribution lies in our empirical study. Using the proposed model, we investigate the implications of macroeconomic uncertainty for the out-of-sample prediction of VaR and ES. It has been well established that macroeconomic volatility and economic activity are important determinants of stock market volatility (Schwert 1989, Engle *et al.* 2013). As volatility is the key input in computing VaR, some papers improve the prediction of VaR using information from the level of economic activity (see, e.g. Opschoor *et al.* 2014, Dominicy and Elst 2015, Massacci 2015, Bauwens *et al.* 2016). Brogaard *et al.* (2020) study the effects of uncertainty on asset prices. They show that increases in political uncertainty cause lower stock returns and higher volatilities. This motivates us to investigate the effect of uncertainty on the conditional distribution of asset returns as expressed by VaR and ES. To the best of our knowledge, we have not found any papers that detect the implications of macroeconomic uncertainty for improving the forecasting accuracy of VaR, let alone for improving ES prediction. We fill this gap using the new model proposed in this paper.

We consider the volatility of two macroeconomic variables: industrial production (IP) and the producer price index (PPI). Our sample data covers the period from January 1928 through December 2018. We forecast the daily VaR and ES of S&P 500 index returns after January 2009 using the newly developed model of GARCH-MIDAS with macroeconomic volatility and FZ loss. The competing models used for comparison are rolling window methods, GAS models under QMLE and FZ minimization estimates, GARCH with various distributions, a hybrid model of PZC, and the corresponding GARCH-MIDAS with QMLE estimates.

Our out-of-sample results indicate that the GARCH-MIDAS model with FZ loss produces more accurate VaR and ES forecasts when incorporating PPI and IP uncertainty as the driver of the long-term volatility component. Based on the model confidence set (MCS) test (Hansen *et al.* 2011), we find that GARCH-MIDAS with FZ minimization estimates provide better forecasts of market risk than their counterpart with QMLE estimates. We provide strong evidence that macroeconomic volatilities are helpful for improving the out-of-sample forecasting performance of VaR and ES.

The remainder of this paper is organized as follows: Section 2 shows the general specification as developed by PZC; Section 3 presents our model and the methodology of VaR and ES estimation; Section 4 and Section 5 report on the simulation and empirical analyses, respectively; and the last section concludes the paper.

## 2. Specification

### 2.1. The Fissler and Ziegel (2016) loss function: A brief review

Fissler and Ziegel (2016) propose a class of loss functions that is consistent for VaR and ES, expressed as

$$\begin{aligned} L_{FZ}(r, v, e; \alpha_{level}, G_1, G_2) \\ = (1(r \leq v) - \alpha_{level}) \left( G_1(v) - G_1(r) + \frac{1}{\alpha_{level}} G_2(e)v \right) \\ - G_2(e) \left( \frac{1}{\alpha_{level}} 1(r \leq v)r - e \right) \\ - \int_{-e}^{\infty} G_2(x) dx \end{aligned} \quad (1)$$

where  $r$  is the stochastic return on the risky asset (e.g. stock return).  $v$  and  $e$  are the VaR and ES of the risky asset given  $\alpha_{level}$  level, respectively.  $1(A)$  denotes the indicator function, having the value 1 when the event  $A$  occurs and 0 otherwise.  $G_1(\cdot)$  is weakly increasing, and  $G_2(\cdot)$  is also weakly increasing as well as strictly positive. The above class is referred to as the FZ loss function. As Fissler and Ziegel (2016) show,  $VaR_t$  and  $ES_t$  are jointly estimated by minimizing any loss functions in equation (1):

$$\begin{aligned} (VaR_{i,t}, ES_{i,t}) = \arg \min_{v,e} E_{i-1,t}[L_{FZ}(r_{i,t}, v, e; \alpha_{level}, G_1, G_2)], \\ \forall i = 1, \dots, N_t, \end{aligned} \quad (2)$$

where  $r_{i,t}$  is the returns on the high-frequency  $i$  (e.g. day) in the low-frequency  $t$  (e.g. month).  $N_t$  denotes the number of days within the period  $t$ . Broadly speaking, there are numerous loss functions that can meet the conditions of  $G_1$  and  $G_2$ . In practice, we therefore select some useful functions to achieve a desired property, such as the loss differences being homogeneous of degree zero, which can give greater power to future volatility forecasts (see, e.g. Patton and Shepard 2009, Nolde and Ziegel 2017). Guided by Proposition 1 in Patton *et al.* (2019), we set  $G_1(x) = 0$  and  $G_2(x) = -1/x$ ,

and the difference of general loss function in equation (1) is zero-homogeneous with a parsimonious form:

$$L_{FZ0}(r, v, e; \alpha_{level}) = -\frac{1}{\alpha_{level}e} 1(r \leq v)(v - r) + \frac{v}{e} + \ln(-e) - 1. \quad (3)$$

As shown by Patton *et al.* (2019), the  $L_{FZ0}$  has nice behaviors, such as the implied VaR ( $v$ ) and implied ES ( $e$ ) loss functions resembling the ‘tick’ and ‘QLIKE’ loss function, respectively.

### 3. Component volatility model and VaR/ES estimation

#### 3.1. The new specification and estimation

Naturally, the magnitude of risk measure should change along with the change in risky asset returns. It is thus reasonable to allow the VaR ( $v$ ) and ES ( $e$ ) to be time-varying. Building on Patton *et al.* (2019), the VaR ( $v$ ) and ES ( $e$ ) is dynamic and the parametric function of  $r_{i-1,t}$  and  $Z_{t-1}$  conditional on the information set  $\mathcal{F}_{i-1,t}$  up to time  $i-1$  of period  $t$ . As usual, the information set is a  $\sigma$ -field generated by observable variables. That is  $\mathcal{F}_{i-1,t} = \sigma\{r_{i-1,t}, r_{i-2,t}, \dots, r_{1,t}, Z_{t-1}, Z_{t-2}, \dots, Z_1\}$ :

$$\begin{aligned} & (VaR_{\alpha_{level}}(r_{i,t}|\mathcal{F}_{i-1,t}), ES_{\alpha_{level}}(r_{i,t}|\mathcal{F}_{i-1,t})) \\ &= (v(r_{i-1,t}, Z_{t-1}; \Theta), e(r_{i-1,t}, Z_{t-1}; \Theta)), \end{aligned} \quad (4)$$

where  $Z_{t-1}$  is the exogenous low-frequency variable at time  $t-1$ , which may contain some useful macroeconomic information. For clarity, we allow variables  $r_{i,t}$  and  $Z_t$  to have different sampling frequencies.  $\Theta$  denotes the parameter set collecting all of the parameters in our model and can be estimated through

$$\begin{aligned} \hat{\Theta} = \arg \min_{\Theta} \frac{1}{N_T} \sum_{t=1}^T \sum_{i=1}^{N_t} L_{FZ0}(r_{i,t}, v(r_{i-1,t}, Z_{t-1}; \Theta), \\ e(r_{i-1,t}, Z_{t-1}; \Theta); \alpha_{level}) \end{aligned} \quad (5)$$

where  $N_T = \sum_{t=1}^T N_t$ . The following section establishes the asymptotic properties for our estimators by verifying the assumptions in Patton *et al.* (2019), who use the  $M$ -estimators theory of White (1996) and Newey and McFadden (1994) to obtain the consistency and asymptotic normality for their semiparametric model.

Specifically, the structure of the dynamics of VaR ( $v$ ) and ES( $e$ ) is

$$\begin{aligned} v_i &= a\sqrt{\tau_t \times g_{i,t}}, \quad \text{where } a = F_z^{-1}(\alpha_{level}) \\ e_i &= b\sqrt{\tau_t \times g_{i,t}}, \quad \text{where } b = E[z|z \leq a] \end{aligned} \quad (6)$$

where  $\tau_t$  and  $g_{i,t}$  are the long-run and short-run conditional variance of the asset returns ( $r_{i,t}$ ), respectively.  $F_z^{-1}(\alpha_{level})$  denotes the quantile of stochastic variable  $z$  given the  $\alpha_{level}$  level. For convenience, the VaR ( $v$ ) can also be rewritten as

$$v_i = c \times e_i, \quad (7)$$

where  $c = a/b$ . Engle *et al.* (2013) document that the GARCH-MIDAS model performs better than traditional

volatility models in forecasting. This insight enables us to shed new light on incorporating a component volatility model into the framework of FZ loss minimization.

$$\begin{aligned} r_{i,t} &= \sqrt{\tau_t} \times g_{i,t} z_{i,t}, \\ g_{i,t} &= (1 - \alpha - \beta) + \alpha \frac{r_{i-1,t}^2}{\tau_t} + \beta g_{i-1,t}, \\ \tau_t &= m + \theta \sum_{k=1}^K \psi_k(\omega) Z_{t-k}, \end{aligned} \quad (8)$$

where  $z_{i,t}|\mathcal{F}_{i-1,t} \sim F_z(0, 1)$ . The exogenous variable  $Z_t$  denotes a low-frequency macroeconomic variable. The weighting scheme  $\psi_k$  is specified by the Beta polynomial due to its simplicity and flexibility (Colacito *et al.* 2011):

$$\psi_k(\omega) = \frac{[1 - k/(K+1)]^{\omega-1}}{\sum_{i=1}^K [1 - i/(K+1)]^{\omega-1}}, \quad (9)$$

#### 3.2. The inference

We establish the asymptotic theory for the estimation of a dynamic VaR and ES model by minimizing FZ0 loss function in equation (5). To this end, we require some regularity assumptions.

**ASSUMPTION 1**  $r_{i,t}$  and  $Z_t$  are both strictly stationary processes. The distribution of  $z_{i,t}$  (i.e.  $F_z(\cdot)$ ) behaves well, such as zero mean, unit variance, finite fourth moment, and a unique non-positive  $\alpha$ -quantile. Its density  $f_z$  satisfies  $f_z(\cdot) \leq \tilde{C}$  and  $|f_z(x_1) - f_z(x_2)| \leq \tilde{C}|x_1 - x_2|$  for some constant  $\tilde{C}$ .

**ASSUMPTION 2** Let  $\Theta_0 = [\alpha_0, \beta_0, \theta_0, \omega_0, b_0, c_0] \in \Theta \subset \mathbb{R}^6$  be the true parameter vector that is in the interior of  $\Theta$ , which is a compact and convex parameter space<sup>†</sup>. That is, for any vector  $[\alpha, \beta, \theta, \omega, b, c] \in \Theta$ ,  $\alpha, \beta \in (0, 1)$ ,  $\theta, \omega, c > 0$ ,  $b < 0$ , and  $\alpha^2 E[z_{i-1,t}^4] + 2\alpha\beta + \beta^2 < 1$ . For convenience, we also assume that  $\underline{c}_x \leq x \leq \bar{c}_x$  for some constant  $\underline{c}_x$  and  $\bar{c}_x$  where  $x \in \{\alpha, \beta, \theta, \omega, b, c\}$ <sup>‡</sup>.

**ASSUMPTION 3**  $E|r_{i,t}|^{4+\delta_c} < \infty$ ,  $E|Z_t|^{4+\delta_c} < \infty$  for some constants  $\delta_c > 0$ .

**REMARK 1** Most of these assumptions are standard and can be found in the literature on the asymptotic properties of GARCH-type models (see, e.g. Lumsdaine 1996, Carrasco and Chen 2002, Patton *et al.* 2019). For example, assumption 2 provides a sufficient condition for  $\beta$ -mixing sequences  $\{r_{i,t}, Z_t\}$ , as well as their derivative sequences  $\{v_{i,t}(\Theta), e_{i,t}(\Theta), \nabla' v_{i,t}(\Theta), \nabla' e_{i,t}(\Theta)\}$ , which implies  $\alpha$ -mixing sequences. We therefore establish the asymptotic normality

<sup>†</sup> Newey and McFadden 1994 state that the compactness assumption is often ignored in applications. Although mixing is an appealing dependence concept, it shares with ergodicity the property that it can be somewhat difficult to verify theoretically and is impossible to verify empirically (White 2014).

<sup>‡</sup> We also compute the compactness restriction condition  $\alpha^2 E[z_{i-1,t}^4] + 2\alpha\beta + \beta^2$ , and find that it equals 0.998, 0.995, and 0.951 for GM-IP-FZ, GM-PPI-FZ, and GM-RV-FZ models, respectively. These values also satisfy the conditions in assumption 2.

using the central limit theorem for  $\alpha$ -mixing sequences proposed by Hall and Heyde (2014). Assumption 3 is needed to show the uniform law of large numbers (LLN), which is important to obtain consistency for the estimators.

**THEOREM 1** *Under assumptions 1–3, we have the consistency*

$$\hat{\Theta}_{N_T} \xrightarrow{p} \Theta_0, \quad N_T \rightarrow \infty, \quad (10)$$

and the asymptotic normality

$$\sqrt{N_T} A_0^{-1/2} B_0 (\hat{\Theta}_{N_T} - \Theta_0) \xrightarrow{d} \mathcal{N}(0, I), \quad \text{as } N_T \rightarrow \infty, \quad (11)$$

where  $I$  refers to  $(6 \times 6)$  identity matrix, and

$$\begin{aligned} B_0 &= E \left[ \begin{aligned} & \frac{f_t(v_{i,t}(\Theta_0) | \mathcal{F}_{i-1,t})}{-e_{i,t}(\Theta_0) \alpha_{level}} \nabla' v_{i,t}(\Theta_0) \nabla v_{i,t}(\Theta_0) \\ & + \frac{1}{e_{i,t}(\Theta_0)^2} \nabla' e_{i,t}(\Theta_0) \nabla e_{i,t}(\Theta_0) \end{aligned} \right] \\ A_0 &= E[g_{i,t}(\Theta_0) g_{i,t}(\Theta_0)'] \\ g_{i,t}(\Theta) &= \frac{\partial L(r_{i,t}, v_{i-1,t}(\Theta), e_{i-1,t}(\Theta); \alpha)}{\partial \Theta} \\ &= \nabla' v_{i-1,t}(\Theta) \frac{1}{-e_{i-1,t}(\Theta)} \\ &\quad \times \left( \frac{1}{\alpha_{level}} 1(r_{i,t} \leq v_{i-1,t}(\Theta)) - 1 \right) \\ &\quad + \nabla' e_{i-1,t}(\Theta) \frac{1}{e_{i-1,t}^2} \left( \frac{1}{\alpha_{level}} 1(r_{i,t} \leq v_{i-1,t}(\Theta)) \right. \\ &\quad \left. \times (v_{i-1,t} - r_{i,t}) - v_{i-1,t}(\Theta) + e_{i-1,t}(\Theta) \right) \end{aligned}$$

The proof is provided in Appendix A, which builds on Patton *et al.* (2019) by checking whether their assumptions 1 and 2 are satisfied. We also provide the estimates of asymptotic covariance of  $\hat{\Theta}_{N_T}$ , which makes the computation of standard errors and confidence intervals feasible. Our specification can fall into their framework using the chain rule to the derivative of our new model. Thus, we have  $\hat{A}_{N_T} - A_0 \xrightarrow{p} 0$  and  $\hat{B}_{N_T} - B_0 \xrightarrow{p} 0$  under the assumption 3 in Patton *et al.* (2019), where

$$\begin{aligned} \hat{A}_{N_T} &= \frac{1}{N_T} \sum_{t=1}^T \sum_{i=1}^{N_t} g_{i,t}(\hat{\Theta}) g_{i,t}(\hat{\Theta})' \\ \hat{B}_{N_T} &= \frac{1}{N_T} \sum_{t=1}^T \sum_{i=1}^{N_t} \left\{ \frac{1}{2c_T} 1(|r_{i,t} - v_{i,t}(\hat{\Theta})| < c_T) \right. \\ &\quad \times \frac{\nabla' v_{i-1,t}(\hat{\Theta}) \nabla v_{i-1,t}(\hat{\Theta})}{-\alpha_{level} e_{i-1,t}(\hat{\Theta})} + \frac{\nabla' e_{i-1,t}(\hat{\Theta}) \nabla e_{i-1,t}(\hat{\Theta})}{e_{i-1,t}^2(\hat{\Theta})} \left. \right\}. \end{aligned} \quad (12)$$

#### 4. Simulation experiment

To investigate the finite-sample performance of our model, we design a simulation experiment with the DGP using the

GARCH-MIDAS with monthly realized volatility and the estimation using the  $FZ$  loss minimization. The DGP is given as

$$\begin{aligned} r_{i,t} &= \sqrt{\tau_t} \times g_{i,t} z_{i,t}, \\ g_{i,t} &= (1 - \alpha - \beta) + \alpha \frac{r_{i-1,t}^2}{\tau_t} + \beta g_{i-1,t}, \\ \tau_t &= m + \theta \sum_{k=1}^K \psi_k(\omega) RV_{t-k}, \end{aligned} \quad (13)$$

where  $RV_t = \sum_{i=1}^{N_t} r_{i,t}^2$ . As guided by the empirical results in Engle *et al.* (2013) and Patton *et al.* (2019), the true parameters  $(\alpha, \beta, m, \theta, \omega)$  of this DGP are chosen as (0.10944, 0.85004, 0.00003, 0.01054, 6.62953), and the number of lags ( $K$ ) is fixed at 48 months. The distribution  $z_{i,t}$  has two specifications: one is the standard normal distribution and the other is the skewed  $t$  distribution of Hansen (1994) for a tougher environment with the degree of freedom and skewness parameters being 5 and  $-0.5$ , respectively.

We consider a sample of size  $T = 2500$  corresponding to 10 years of daily observations and repeat all simulations 1000 times. As stated in Lemma 3 below,  $m$  is not identified in our specification with  $FZ$  loss minimization. To avoid this issue, we set  $m$  to the true value (0.00003) in simulate exercise and the counterpart estimates using GARCH-MIDAS model with QMLE in empirical work, respectively. Table 1 presents the results for the estimates of GARCH-MIDAS obtained by the (quasi-) maximum likelihood method (QMLE), and table 2 presents the corresponding results for the estimates obtained by the  $FZ$  minimization method. The left panel of each table reports the results for normally distributed innovations. The QMLE estimates of  $\alpha, \beta$  and  $\theta$  have medians close to the true ones and average biases close to zero, suggesting that the parameter estimates are centered on the true values. The QMLE estimates of these three parameters have lower average biases and standard deviations than the  $FZ$  minimization estimates, suggesting better finite-sample properties. This result is expected. However, the QMLE estimate of  $\omega$  is more volatile than the corresponding estimate based on  $FZ$  minimization.<sup>†</sup>

We compute the coverage probabilities for 95% confidence intervals for each parameter. The coverage rates are constructed based on the estimates of standard errors, using Theorem 1 with bandwidth  $c_T = \lfloor T^{-1/3} \rfloor$ . For a normal distribution, the coverage probabilities of the three parameter estimates  $\alpha, \beta$  and  $\theta$  are between 0.891 and 0.964, depending on the choice of estimation methodology. However, the coverage probabilities of  $\omega$  are quite low, regardless of which estimation method is applied. Therefore, some caution is required when interpreting standard errors for this parameter in the GARCH-MIDAS model.<sup>‡</sup> The coverage probabilities

<sup>†</sup> The estimate of  $\omega$  is much more volatile than all the other parameter estimates under both QMLE and  $FZ$  minimization frameworks in terms of Std Dev and MSE.

<sup>‡</sup> The Lemma 3 shows that our specification is identified under the restriction for normalized parameter  $m$  and the weighting scheme  $\psi_k(\omega) = 1$ , but some parameters could be not identified well in practice due to the sophisticated specification.



Table 1. Simulation results obtained by (Quasi-) maximum likelihood method.

|          | DGP: Normal distribution |         |          |          | DGP: Skewed $t$ distribution |         |          |          |
|----------|--------------------------|---------|----------|----------|------------------------------|---------|----------|----------|
|          | $\alpha$                 | $\beta$ | $\theta$ | $\omega$ | $\alpha$                     | $\beta$ | $\theta$ | $\omega$ |
| True     | 0.109                    | 0.850   | 0.011    | 6.630    | 0.109                        | 0.850   | 0.011    | 6.630    |
| Median   | 0.110                    | 0.847   | 0.010    | 2.203    | 0.111                        | 0.846   | 0.009    | 3.738    |
| Avg bias | 0.000                    | -0.003  | 0.000    | -1.766   | 0.005                        | -0.009  | 0.008    | 0.066    |
| MSE      | 0.000                    | 0.001   | 0.000    | 33.995   | 0.001                        | 0.002   | 0.008    | 48.086   |
| St dev   | 0.015                    | 0.022   | 0.005    | 5.559    | 0.033                        | 0.045   | 0.088    | 6.938    |
| Coverage | 0.947                    | 0.958   | 0.920    | 0.449    | 0.904                        | 0.889   | 0.834    | 0.568    |
| Accuracy | —                        | —       | —        | —        | 2.186                        | 2.072   | 17.593   | 1.248    |

Notes: This table provides the estimation results using Quasi-maximum likelihood method under the two data generating processes (DGPs): GARCH-MIDAS with realized volatility for standard normal and skewed  $t$  distributions. The first row shows the true values of the parameters. The median, the average bias, the mean squared error, the standard deviation, and the coverage rates for 95% of estimated parameters are presented in the second, third, fourth, fifth, and sixth rows, respectively. The last row calculates the relative accuracy of the estimates measured by the ratio between the standard deviation of parameter estimates under the skewed  $t$  distribution and that of parameter estimates under the standard normal distribution. All results are based on 1000 Monte Carlo simulations with sample size  $T = 2500$ .

Table 2. Simulation results obtained by FZ loss minimization with normal distribution.

|                                                 | DGP: Normal distribution |         |          |          | DGP: Skewed $t$ distribution |         |          |          |
|-------------------------------------------------|--------------------------|---------|----------|----------|------------------------------|---------|----------|----------|
|                                                 | $\alpha$                 | $\beta$ | $\theta$ | $\omega$ | $\alpha$                     | $\beta$ | $\theta$ | $\omega$ |
| The probability level: $\alpha_{level} = 1\%$   |                          |         |          |          |                              |         |          |          |
| True                                            | 0.109                    | 0.850   | 0.011    | 6.630    | 0.109                        | 0.850   | 0.011    | 6.630    |
| Median                                          | 0.120                    | 0.838   | 0.019    | 3.400    | 0.122                        | 0.830   | 0.014    | 3.632    |
| Avg bias                                        | 0.014                    | -0.018  | 0.024    | -2.532   | 0.026                        | -0.045  | 0.052    | -2.325   |
| MSE                                             | 0.002                    | 0.003   | 0.004    | 14.804   | 0.006                        | 0.014   | 0.023    | 14.422   |
| St dev                                          | 0.037                    | 0.051   | 0.057    | 2.899    | 0.072                        | 0.110   | 0.143    | 3.004    |
| Coverage                                        | 0.891                    | 0.932   | 0.964    | 0.802    | 0.900                        | 0.892   | 0.967    | 0.869    |
| Accuracy                                        | —                        | —       | —        | —        | 1.940                        | 2.166   | 2.511    | 1.036    |
| The probability level: $\alpha_{level} = 2.5\%$ |                          |         |          |          |                              |         |          |          |
| True                                            | 0.109                    | 0.850   | 0.011    | 6.630    | 0.109                        | 0.850   | 0.011    | 6.630    |
| Median                                          | 0.117                    | 0.843   | 0.017    | 3.168    | 0.119                        | 0.836   | 0.015    | 3.365    |
| Avg bias                                        | 0.011                    | -0.013  | 0.025    | -2.595   | 0.020                        | -0.030  | 0.047    | -2.427   |
| MSE                                             | 0.001                    | 0.002   | 0.006    | 15.353   | 0.004                        | 0.008   | 0.026    | 15.150   |
| St dev                                          | 0.033                    | 0.044   | 0.070    | 2.937    | 0.057                        | 0.083   | 0.153    | 3.045    |
| Coverage                                        | 0.917                    | 0.947   | 0.953    | 0.755    | 0.882                        | 0.902   | 0.949    | 0.793    |
| Accuracy                                        | —                        | —       | —        | —        | 1.737                        | 1.886   | 2.199    | 1.037    |
| The probability level: $\alpha_{level} = 5\%$   |                          |         |          |          |                              |         |          |          |
| True                                            | 0.109                    | 0.850   | 0.011    | 6.630    | 0.109                        | 0.850   | 0.011    | 6.630    |
| Median                                          | 0.117                    | 0.844   | 0.017    | 2.947    | 0.119                        | 0.834   | 0.014    | 3.389    |
| Avg bias                                        | 0.009                    | -0.010  | 0.025    | -2.726   | 0.015                        | -0.025  | 0.035    | -2.388   |
| MSE                                             | 0.001                    | 0.002   | 0.005    | 16.048   | 0.002                        | 0.005   | 0.016    | 15.082   |
| St dev                                          | 0.029                    | 0.038   | 0.065    | 2.937    | 0.045                        | 0.067   | 0.123    | 3.064    |
| Coverage                                        | 0.920                    | 0.946   | 0.953    | 0.725    | 0.937                        | 0.958   | 0.955    | 0.801    |
| Accuracy                                        | —                        | —       | —        | —        | 1.550                        | 1.769   | 1.898    | 1.044    |
| The probability level: $\alpha_{level} = 10\%$  |                          |         |          |          |                              |         |          |          |
| True                                            | 0.109                    | 0.850   | 0.011    | 6.630    | 0.109                        | 0.850   | 0.011    | 6.630    |
| Median                                          | 0.115                    | 0.845   | 0.016    | 2.826    | 0.115                        | 0.839   | 0.013    | 3.202    |
| Avg bias                                        | 0.009                    | -0.009  | 0.030    | -2.752   | 0.013                        | -0.019  | 0.037    | -2.503   |
| MSE                                             | 0.001                    | 0.002   | 0.006    | 16.660   | 0.002                        | 0.004   | 0.020    | 15.727   |
| St dev                                          | 0.031                    | 0.039   | 0.074    | 3.016    | 0.044                        | 0.059   | 0.135    | 3.077    |
| Coverage                                        | 0.911                    | 0.948   | 0.942    | 0.694    | 0.927                        | 0.937   | 0.929    | 0.707    |
| Accuracy                                        | —                        | —       | —        | —        | 1.419                        | 1.502   | 1.810    | 1.020    |

Notes: This table provides the estimation results using the minimization of FZ loss function under the two data generating processes (DGPs): GARCH-MIDAS with realized volatility for standard normal and skewed  $t$  distributions. The first row shows the true values of the parameters. The median, the average bias, the mean squared error, the standard deviation, and the coverage rates for 95% of estimated parameters are presented in the second, third, fourth, fifth, and sixth rows, respectively. The last row calculates the relative accuracy of the estimates measured by the ratio between the standard deviation of parameter estimates under the skewed  $t$  distribution and that of parameter estimates under the standard normal distribution. All results are based on 1000 Monte Carlo simulations with sample size  $T = 2500$ .

of  $\alpha$  and  $\beta$  based on the QMLE estimate are closer to 95% than their counterparts based on  $FZ$  minimization, while the relative performance of the other two parameters are the reverse.

The right panel of each table reports the results for skew- $t$  distribution. We find that the coverage probabilities of almost all parameters based on  $FZ$  minimization are closer to 95% than their counterparts based on QMLE estimation. This situation is not affected by the changes in quantile  $\alpha_{level}$  for the  $FZ$  method to estimate parameters.

We also compute the inaccuracy rate, which is defined as the ratio between the standard deviation of parameter estimates under the skew- $t$  distribution and that of parameter estimates under the standard normal distribution. Intuitively, a higher inaccuracy rate implies that the corresponding parameter estimate is more sensitive to the change of innovation distribution. We find that for all parameters, the  $FZ$  minimization estimates have a lower inaccuracy rate than the corresponding QMLE estimates. That is, the  $FZ$  minimization estimation is more robust to model misspecification.

## 5. Empirical results for ES and VaR

### 5.1. The data

We collect the daily returns on the S&P 500 index from Bloomberg, covering the period from January 3, 1928 through December 31, 2018, which provides 22557 observations. The monthly data for IP and PPI are obtained from the Federal Reserve Bank of Saint Louis<sup>†</sup>, providing 1092 observations. We use three variables to capture macroeconomic uncertainty. The first is monthly realized volatility (RV), defined as the sum of squared daily returns in each month, given by

$$RV_t = \sum_{i=1}^{N_t} r_{i,t}^2, \quad (14)$$

where  $r_{i,t}$  denotes the stock returns on the  $i$ th day of month  $t$ . The other two variables reflecting uncertainty are the IP and PPI volatilities. Following Engle *et al.* (2013), we construct the macroeconomic volatility as follows:

$$X_t = \sum_{j=1}^{12} \rho_j D_{j,t} + \sum_{j=1}^{12} \vartheta_j X_{t-j} + \epsilon_t, \quad (15)$$

where  $X_t$  is a macroeconomic level variable of interest and  $D_{j,t}$  is a monthly dummy variable. The volatility of  $X_t$  can be proxied by the squared residuals (i.e.  $\epsilon_t^2$ ). Figure 1 plots the log changes of these economic variables (i.e. returns on stock index) and their volatilities during the whole sample period. The summary statistics for stock returns and macroeconomic volatilities are reported in table 3.

### 5.2. Competing models

We use four types of models to produce VaR and ES forecasts. The first type contains three rolling window methods

with window lengths of 125, 250 and 500 days. The second type of model is GARCH with different distributions based on QMLE estimates. The GARCH specification is given by

$$r_t = \sigma_t \eta_t, \quad \eta_t \sim i.i.d F_\eta(0, 1) \\ \sigma_t^2 = \bar{\omega} + \beta \sigma_{t-1}^2 + \alpha r_{t-1}^2, \quad (16)$$

where  $F_\eta(0, 1)$  is a strictly increasing distribution function with mean zero and variance one. We consider three distributions: normal, Skew- $t$ , and empirical distributions. The VaR and ES from the GARCH model can be written as

$$v_t = a \cdot \sigma_t, \quad \text{where } a = F_\eta^{-1}(\alpha_{level}) \\ e_t = b \cdot \sigma_t, \quad \text{where } b = E[\eta | \eta \leq a]. \quad (17)$$

The third type of model includes those used in the PZC paper. Besides the GARCH based on  $FZ$  minimization estimates, we also use the two-factor GAS model suggested by PZC:

$$\begin{bmatrix} v_{t+1} \\ e_{t+1} \end{bmatrix} = \mathbf{W} + \mathbf{B} \begin{bmatrix} v_t \\ e_t \end{bmatrix} + \mathbf{A} H_t^{-1} \nabla_t \quad (18)$$

where  $\mathbf{W}$  is a vector ( $2 \times 1$ ), and both  $\mathbf{B}$  and  $\mathbf{A}$  are matrices ( $2 \times 2$ ). The forcing part is specified as

$$H_t^{-1} \nabla_t = \begin{bmatrix} \frac{1}{\kappa_x} \lambda_{v,t} \\ \frac{1}{\alpha_{level}} (\lambda_{v,t} + \alpha_{level} \lambda_{e,t}) \end{bmatrix}$$

where

$$\kappa_x \equiv x f_\eta(x) \\ \lambda_{v,t} \equiv -v_t (1(r_t \leq v_t) - \alpha_{level}) \\ \lambda_{e,t} \equiv \frac{1}{\alpha_{level}} 1(r_t \leq v_t) r_t - e_t,$$

where  $f_\eta(\cdot)$  denotes a density function of  $\eta$ .

As a special case, the one-factor GAS is given by,

$$v_t = a \exp(\kappa_t) \\ e_t = b \exp(\kappa_t) \\ \kappa_t = \bar{\omega} + \beta \kappa_{t-1} + \alpha H_{t-1}^{-1} s_{t-1} \quad (19)$$

where  $b < a < 0$  and the forcing variable,  $H_{t-1}^{-1} s_{t-1}$  is obtained from the objective loss function ( $FZ0$ ).

PZC suggests a hybrid GAS/GARCH model that presents good out-of-sample performance. This model is,

$$r_t = \exp(\kappa_t) \eta_t, \quad \text{where } \eta_t \sim F_\eta(0, 1) \\ \kappa_t = \bar{\omega} + \beta \kappa_{t-1} + \frac{\alpha}{e_{t-1}} \left( \frac{1}{\alpha_{level}} 1(r_{t-1} \leq v_{t-1}) r_{t-1} - e_{t-1} \right) \\ + \gamma \log(|r_{t-1}|). \quad (20)$$

For the last type of model, we use three variables reflecting macroeconomic uncertainty in GARCH-MIDAS: the realized volatility, IP and PPI volatilities. Based on two different estimation frameworks, we employ a total of 6 GARCH-MIDAS methods. As a summary, 16 econometric models are applied to estimate and forecast VaR and ES. Brief descriptions of these models are provided in table 4.

<sup>†</sup> <https://fred.stlouisfed.org>.



Figure 1. Stock returns, volatilities and macroeconomic variables. This figure plots the returns (daily) on S&P 500 index and its realized variance (monthly), the log changes of industrial production (monthly) and its volatility (monthly), the log changes of producer price index (monthly) and its volatility (monthly) over the period from January 1928 to December 2018.

Table 3. Summary statistics.

|                        | Returns | RV     | IP<br>volatility | PPI<br>volatility |
|------------------------|---------|--------|------------------|-------------------|
| Mean( $\times 100$ )   | 0.022   | 0.288  | 0.021            | 0.009             |
| StdDev( $\times 100$ ) | 1.181   | 0.581  | 0.088            | 0.040             |
| Skewness               | -0.435  | 6.410  | 11.234           | 15.358            |
| Kurtosis               | 21.678  | 59.959 | 172.563          | 308.761           |

Notes: This table provides the summary statistics for returns on S&P 500 index, realized volatility, and the volatilities of industrial production and producer price index.

### 5.3. In sample analysis

For the GARCH-MIDAS model, we need to determine the maximum lag length in the MIDAS regression for the long-term volatility component (8) (i.e. the value of  $K$ ). We let

$K$  change from 12 to 120 months with a fixed step of 12 months, and calculate the  $FZ$  loss for each  $K$  value. Figure 2 shows that the GARCH-MIDAS models with various macroeconomic uncertainty variables all have the lowest  $FZ$  losses at the 5% tail when  $K = 36$ . We therefore choose  $K = 36$  in the following empirical analysis. Figures 3 and 4 plot the VaR and ES series, respectively, at 1%, 2.5%, 5% and 10% tails implied by the GARCH-MIDAS models.

Table 5 presents the estimation results for the GARCH and GARCH-MIDAS models with skew- $t$  distribution by maximum likelihood (ML) method. We observe that the degrees of freedom parameter  $\nu$  is greater than 5 for all four models, which implies that assumption 3 is supported by the empirical data. The original data is skewed to the left with the negative skewness parameter  $\lambda$ . This is consistent with the finding in table 3.

Table 4. The description of competing models.

| Model number | Description                                                                   | Abbreviation |
|--------------|-------------------------------------------------------------------------------|--------------|
| 1            | 125-day rolling window                                                        | RW125        |
| 2            | 250-day rolling window                                                        | RW250        |
| 3            | 500-day rolling window                                                        | RW500        |
| 4            | GARCH model with normal distribution, QMLE estimates                          | GCH-N-MLE    |
| 5            | GARCH model with skew-t distribution, QMLE estimates                          | GCH-SKT-MLE  |
| 6            | GARCH model with empirical distribution function, QMLE estimates              | GCH-EDF-MLE  |
| 7            | Two-factor GAS model, FZ minimization estimates                               | G2-FZ        |
| 8            | One-factor GAS model, FZ minimization estimates                               | G1-FZ        |
| 9            | GARCH model with normal distribution, FZ minimization estimates               | GCH-N-FZ     |
| 10           | The hybrid model of Patton, Ziegel, and Chen (2019)                           | Hybrid-FZ    |
| 11           | GARCH-MIDAS with industrial production uncertainty, QMLE estimates            | GM-IP-MLE    |
| 12           | GARCH-MIDAS with producer price index uncertainty, QMLE estimates             | GM-PPI-MLE   |
| 13           | GARCH-MIDAS with realized volatility, QMLE estimates                          | GM-RV-MLE    |
| 14           | GARCH-MIDAS with industrial production uncertainty, FZ minimization estimates | GM-IP-FZ     |
| 15           | GARCH-MIDAS with producer price index uncertainty, FZ minimization estimates  | GM-PPI-FZ    |
| 16           | GARCH-MIDAS with realized volatility, FZ minimization estimates               | GM-RV-FZ     |

Notes: This table lists the names of econometric models used in the paper.

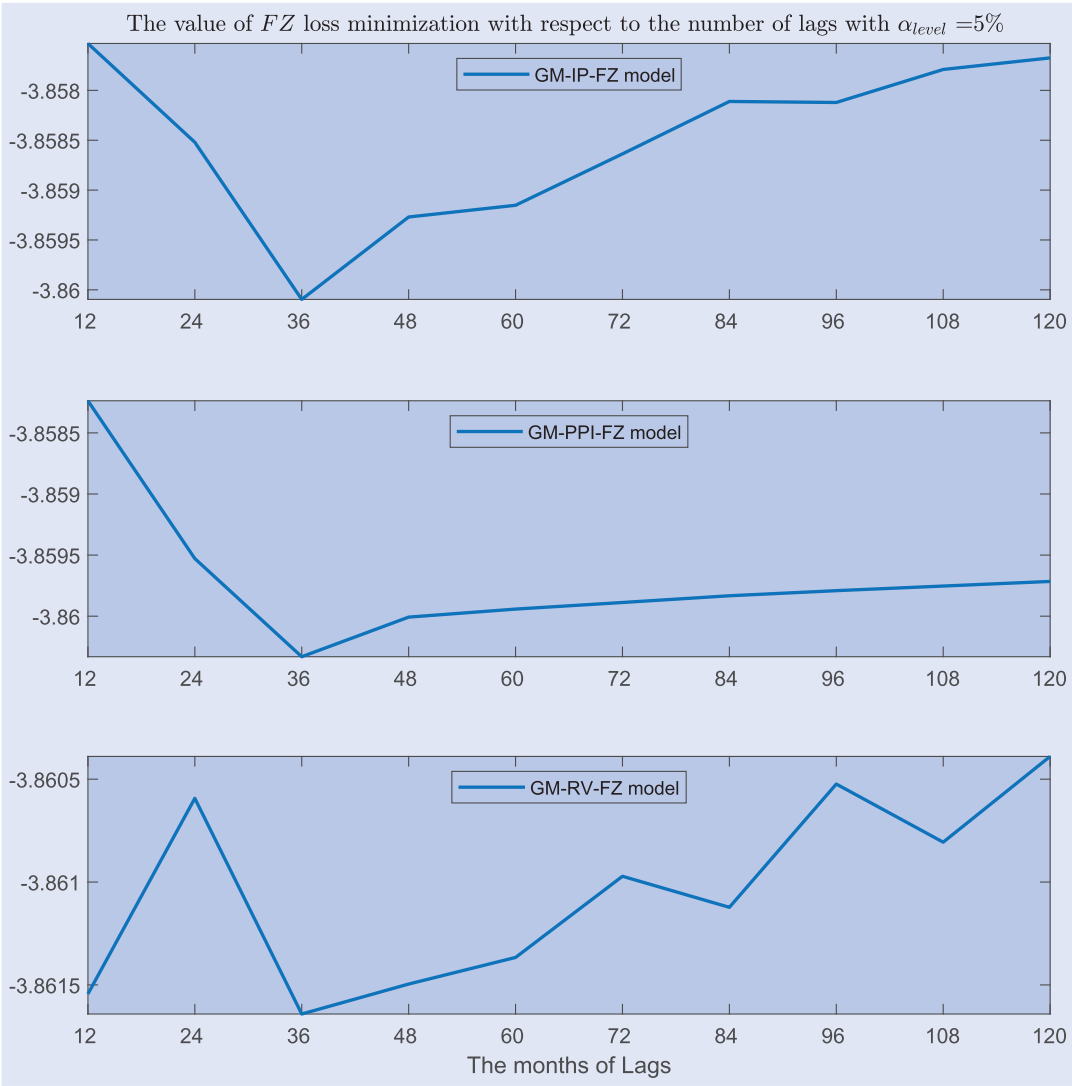


Figure 2. FZ loss function respect to the lag lengths. This figure plots the value of FZ loss minimization with respect to the number of lags based on GM-IP-FZ model, GM-PPI-FZ model, and GM-RV-FZ model with  $\alpha_{level} = 5\%$  over the period from January 1928 to December 2018.



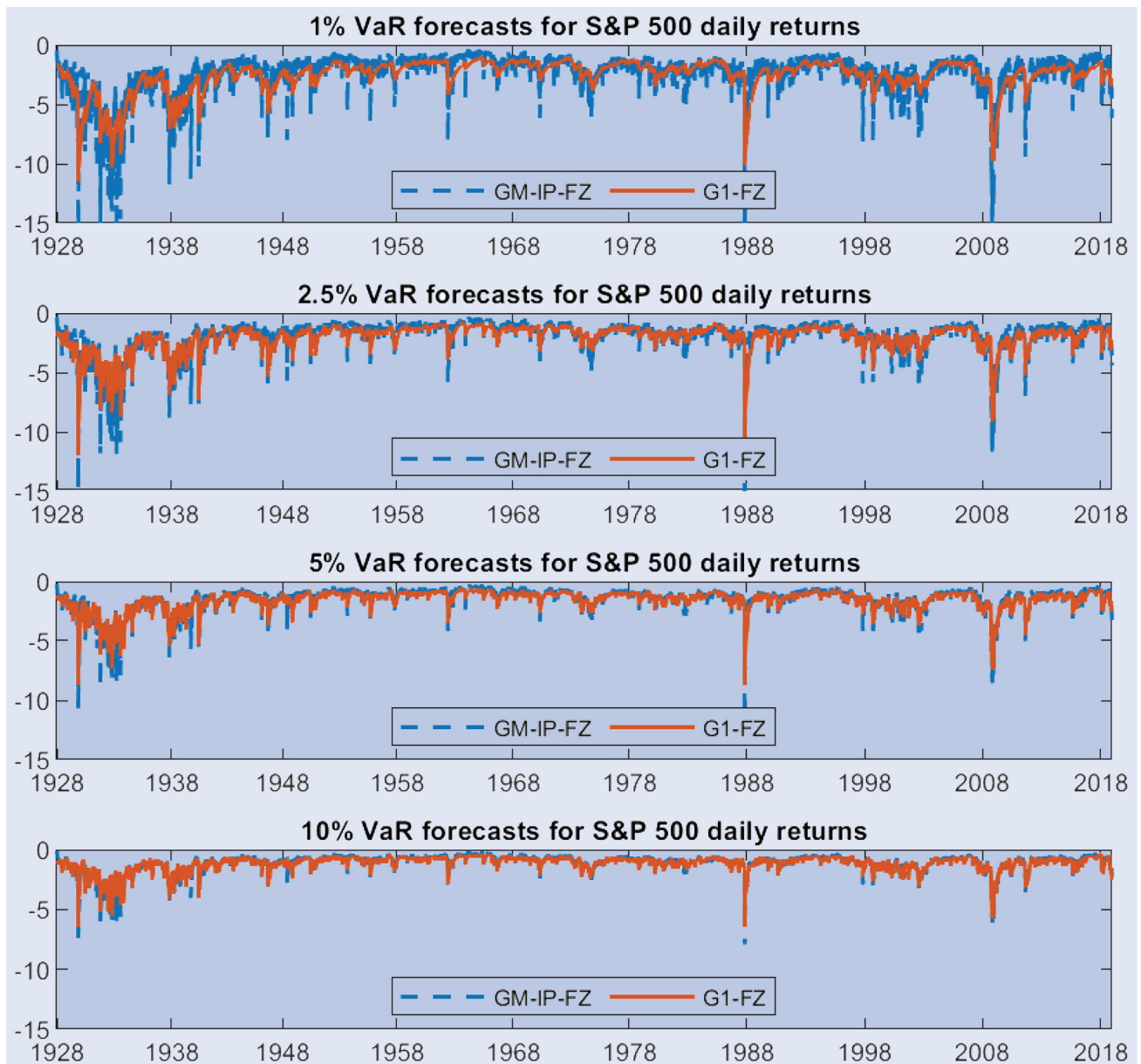


Figure 3. VaR estimates. This figure plots the VaR at 1%, 2.5%, 5%, and 10% level over the period from January 1928 to December 2018.

Table 6 reports the in-sample estimation results for the GARCH-MIDAS models based on *FZ* minimization at the tail  $\alpha_{level} = 5\%$ . We find that the estimates of  $\beta$  change between 0.826 and 0.898, suggesting strong persistence in the short-term volatility component. The estimates of  $\theta$  are significantly positive at the 1% level, indicating that all three uncertainty variables have significant impacts on the long-term volatility component of stock index returns. The estimates of  $b$  are significantly negative and between  $-2.310$  and  $-2.208$ , implying a negative relationship between conditional volatility and ES. The estimates of  $c$  range between 0.705 and 0.708. That is, the positive relationship between VaR and ES at the 5% tail is robust to the choice of macroeconomic uncertainty. We also compute the compactness restriction condition  $\alpha^2 E[z_{i-1,t}^4] + 2\alpha\beta + \beta^2$ , and find that it equals 0.998, 0.995, and 0.951 for GM-IP-FZ, GM-PPI-FZ, and GM-RV-FZ models, respectively. These values also satisfy the conditions in assumption 2.

In table 7 we look at the in-sample average losses of these models across four levels of  $\alpha_{level}$ . The lowest values in each column are preferred and highlighted in bolds. We find that the GARCH-MIDAS with producer price index uncertainty, labeled as GM-PPI-FZ, has the best performance for  $\alpha_{level} = 1\%$  and  $\alpha_{level} = 2.5\%$ , and GARCH-MIDAS with realized volatility, labeled as GM-RV-FZ, has the best performance for  $\alpha_{level} = 5\%$ . This reveals that the GARCH-MIDAS with uncertainty variables is a good alternative to model and forecasting extreme circumstance.

To further compare the models of Patton *et al.* (2019) with our proposed models, table 8 provides Diebold and Mariano (1995) *t*-statistic on the loss difference between these pairwise models across four values of  $\alpha_{level}$ . A positive value means the column model has lower average loss than the row model and so the column model is favored. Values greater than 1.65, 1.96 and 2.58 in absolute imply that the difference is significantly different from zero at 10%, 5% and 1%

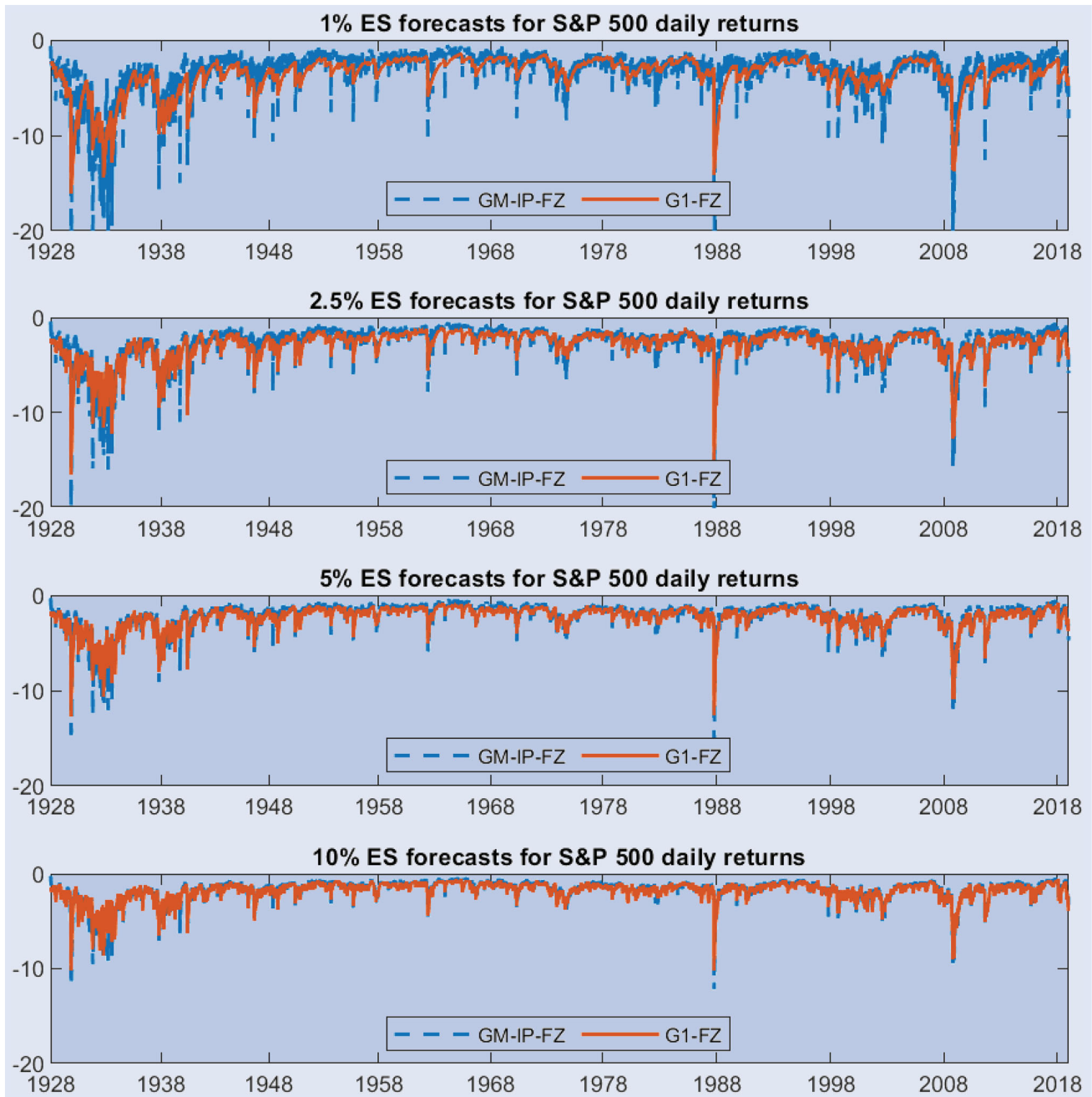


Figure 4. ES estimates. This figure plots the ES at 1%, 2.5%, 5%, and 10% level over the period from January 1928 to December 2018.

significance levels when two-sided test is employed. Consistent with our expectation, the GM-PPI-FZ column contains all positive element for  $\alpha_{level} = 1\%$  and  $\alpha_{level} = 2.5\%$ , reflecting that this model is preferred to all competing models. The gains are almost significant at 5% level with the Diebold and Mariano (1995) statistics greater than 1.96 except Hybrid-FZ case at  $\alpha_{level} = 2.5\%$  with the test statistic of 1.670, significant at 10% level. Again, our models perform better than Patton *et al.* (2019)'s specifications for extreme observations. We will have a discussion about this in section 5.5.

#### 5.4. Out-of-sample performance

This section reports the out-of-sample performance in forecasting market risk during the recent ten years from January

2009 to December 2018. We estimate these models using the period before 2009 as our in-sample period. The parameters are estimated every year using all available data. For example, we obtain the parameters using data from January 1928 to December 2008, and compute all forecasts in 2009 using these parameters. Then, we estimate the parameters again using data from January 1928 to December 2009, and calculate all forecasts in 2010 using these updated parameters.

Panel A of table 9 shows the average losses of volatility models, using the FZ0 loss function. We find that GARCH and GARCH-MIDAS based on QMLE estimates result in greater forecasting losses than their counterparts based on FZ minimization estimates. This is consistent with the result in PZC (2019), which shows that FZ minimization estimation leads to more accurate out-of-sample forecasts. The GARCH-MIDAS model with macroeconomic volatility variables has

Table 5. Estimation results.

|           | GARCH                | GM-IP-MLE            | GM-PPI-MLE           | GM-RV-MLE            |
|-----------|----------------------|----------------------|----------------------|----------------------|
| $\mu$     | 0.000***<br>(0.000)  | 0.000***<br>(0.000)  | 0.000***<br>(0.000)  | 0.000***<br>(0.000)  |
| $\alpha$  | 0.088***<br>(0.009)  | 0.077***<br>(0.009)  | 0.076***<br>(0.011)  | 0.080***<br>(0.028)  |
| $\beta$   | 0.909***<br>(0.010)  | 0.918***<br>(0.010)  | 0.918***<br>(0.011)  | 0.914***<br>(0.024)  |
| $m$       | 0.000***<br>(0.000)  | 0.000***<br>(0.000)  | 0.000***<br>(0.000)  | 0.000<br>(0.000)     |
| $\theta$  |                      | 0.008***<br>(0.000)  | 0.009***<br>(0.003)  | 0.003<br>(0.022)     |
| $\omega$  |                      | 12.574***<br>(2.418) | 12.707***<br>(0.860) | 8.965<br>(14.638)    |
| $\nu$     | 6.431***<br>(0.283)  | 6.659***<br>(0.298)  | 6.659***<br>(0.298)  | 6.511***<br>(0.987)  |
| $\lambda$ | -0.078***<br>(0.008) | -0.080***<br>(0.009) | -0.080***<br>(0.009) | -0.079***<br>(0.021) |

Notes: This table provides the estimation results of GARCH and GARCH-MIDAS models with skew- $t$  distribution by QMLE over the period from January 1928 to December 2018. The standard errors are reported in parentheses. The asterisks \*, \*\*, and \*\*\* indicate significance at the 10%, 5%, and 1% levels, respectively.

Table 6. Estimation results based on FZ loss minimization ( $\alpha_{level} = 5\%$ ).

|          | GM-IP-FZ             | GM-PPI-FZ            | GM-RV-FZ             |
|----------|----------------------|----------------------|----------------------|
| $\alpha$ | 0.093***<br>(0.007)  | 0.094***<br>(0.008)  | 0.128***<br>(0.013)  |
| $\beta$  | 0.898***<br>(0.007)  | 0.894***<br>(0.009)  | 0.826***<br>(0.017)  |
| $\theta$ | 0.366***<br>(0.040)  | 0.781***<br>(0.083)  | 0.051***<br>(0.008)  |
| $\omega$ | 1.010***<br>(0.043)  | 1.478***<br>(0.223)  | 2.044***<br>(0.189)  |
| $b$      | -2.301***<br>(0.045) | -2.310***<br>(0.050) | -2.208***<br>(0.131) |
| $c$      | 0.708***<br>(0.008)  | 0.707***<br>(0.008)  | 0.705***<br>(0.007)  |

Notes: This table provides the estimation results obtained by FZ loss minimization at  $\alpha_{level} = 5\%$  over the period from January 1928 to December 2018. The standard errors are reported in parentheses. The asterisks \*, \*\*, and \*\*\* indicate significance at the 10%, 5%, and 1% levels, respectively.

a lower forecasting loss than the GARCH model with normal distribution. That is, we obtain more accurate VaR and ES forecasts by accounting for the effects of macroeconomic uncertainty. This result holds regardless of whether the estimation method is QMLE or FZ minimization. Observing the forecasting losses, we find that the GM-PPI-FZ performs best for 1% and 2.5% tails and GM-RV-FZ performs best for 5% tail, while for the 10% tail the hybrid model of PZC (2019) works better than the competing models.

As a complement to the comparison of forecasting performance, we use a goodness-of-fit test.

$$\begin{aligned}\lambda_{v,t}^s &= c_v + b_v \lambda_{v,t-1}^s + a_v v_t + z_{v,t} \\ \lambda_{e,t}^s &= c_e + b_e \lambda_{e,t-1}^s + a_e e_t + z_{e,t}\end{aligned}\quad (21)$$

Table 7. In sample (IS) average losses.

|             | 1%           | 2.5%         | 5%           | 10%          |
|-------------|--------------|--------------|--------------|--------------|
| RW125       | 1.344        | 1.040        | 0.811        | 0.564        |
| RW250       | 1.361        | 1.050        | 0.829        | 0.577        |
| RW500       | 1.374        | 1.090        | 0.868        | 0.610        |
| GCH-N-MLE   | 1.247        | 0.960        | 0.753        | 0.526        |
| GCH-SKT-MLE | 1.164        | 0.933        | 0.745        | 0.523        |
| GCH-EDF-MLE | 1.163        | 0.933        | 0.745        | 0.523        |
| G2-FZ       | 1.214        | 0.943        | 0.740        | 0.507        |
| G1-FZ       | 1.213        | 0.943        | 0.742        | 0.512        |
| GCH-N-FZ    | 1.160        | 0.932        | 0.745        | 0.522        |
| Hybrid-FZ   | 1.179        | 0.937        | 0.744        | <b>0.511</b> |
| GM-IP-MLE   | 1.252        | 0.960        | 0.752        | 0.525        |
| GM-PPI-MLE  | 1.238        | 0.952        | 0.747        | 0.522        |
| GM-RV-MLE   | 1.237        | 0.954        | 0.745        | 0.523        |
| GM-IP-FZ    | 1.160        | 0.932        | 0.745        | 0.522        |
| GM-PPI-FZ   | <b>1.148</b> | <b>0.925</b> | 0.740        | 0.520        |
| GM-RV-FZ    | 1.154        | 0.927        | <b>0.736</b> | 0.520        |

Notes: This table provides the average loss based on equation (3), and the  $p$  values from goodness-of-fit (GoF) tests for VaR and ES forecasts. The bold values denote the lowest loss function at a given tail. The evaluation period is from January 1928 to December 2008.

where

$$\begin{aligned}\lambda_{v,t}^s &\equiv \frac{\lambda_{v,t}}{v} = 1(r_t \leq v_t) - \alpha_{level} \\ \lambda_{e,t}^s &\equiv \frac{\lambda_{e,t}}{e_t} = \frac{1}{\alpha_{level}} 1(r_t \leq v_t) \frac{r_t}{e_t} - 1\end{aligned}$$

Based on the simple regressions given in (21), we test the null hypothesis that all the coefficients are zero using the  $F$  statistic.

Panel B and Panel C report the results of goodness-of-fit tests for out-of-sample forecasts of VaR and ES, respectively. These goodness-of-fit tests are actually forecast optimality tests. The  $p$ -values greater than 10% (bolded) indicate that the model passes the test. We find that at a 10% confidence level the hybrid, GM-PPI-FZ and GM-PPI-RV models pass the VaR tests for all tails. A similar result is found for ES tests.

To examine whether the forecasting performance of various models is significantly different, we use the model confidence set test of Hansen *et al.* (2011). The rationale of this test is that the available information is not sufficient to choose a single best model. Alternatively, there is a model pool, namely a model confidence set, which has significantly better out-of-sample performance than the other competitors at a given confidence level. The models within the pool perform equally well in the statistical sense.

Following the notations in Hansen *et al.* (2011), the relative sample loss statistics are defined as  $\bar{d}_{ij} = n^{-1} \sum_{t=1}^n d_{ij,t}$  and  $\bar{d}_i = m^{-1} \sum_{j \in M} \bar{d}_{ij}$ . Here  $d_{ij}$  measures the relative sample loss between the  $i$ th and  $j$ th models, while  $\bar{d}_i$  is the sample loss of the  $i$ th model relative to the average across models in  $M$ . Let  $\mu_i = E(\bar{d}_i)$  and the null hypothesis be  $H_0 : \{\mu_i \leq 0 \text{ for all } i \in M\}$ . To test for the null hypothesis, Hansen *et al.* (2011) construct the following statistic:

$$T_{\max, M} = \max_{i \in M} t_i \quad (22)$$

Table 8. Diebold and Mariano (1995) test for the average in sample (IS) loss differences.

|                          | G2-FZ   | G1-FZ   | GCH-N-FZ | Hybrid-FZ | GM-IP-FZ | GM-PPI-FZ | GM-RV-FZ |
|--------------------------|---------|---------|----------|-----------|----------|-----------|----------|
| $\alpha_{level} = 1\%$   |         |         |          |           |          |           |          |
| G2-FZ                    |         | 0.079   | 3.802    | 3.644     | 3.817    | 4.814     | 3.715    |
| G1-FZ                    | − 0.079 |         | 3.046    | 2.817     | 3.045    | 3.945     | 3.544    |
| GCH-N-FZ                 | − 3.802 | − 3.046 |          | − 1.709   | 0.544    | 2.074     | 0.763    |
| Hybrid-FZ                | − 3.644 | − 2.817 | 1.709    |           | 1.719    | 2.790     | 1.891    |
| GM-IP-FZ                 | − 3.817 | − 3.045 | − 0.544  | − 1.719   |          | 2.038     | 0.736    |
| GM-PPI-FZ                | − 4.814 | − 3.945 | − 2.074  | − 2.790   | − 2.038  |           | − 0.695  |
| GM-RV-FZ                 | − 3.715 | − 3.544 | − 0.763  | − 1.891   | − 0.736  | 0.695     |          |
| $\alpha_{level} = 2.5\%$ |         |         |          |           |          |           |          |
| G2-FZ                    |         | 0.021   | 1.451    | 1.307     | 1.466    | 2.339     | 1.778    |
| G1-FZ                    | − 0.021 |         | 1.314    | 0.837     | 1.326    | 2.132     | 1.665    |
| GCH-N-FZ                 | − 1.451 | − 1.314 |          | − 0.707   | 0.881    | 2.282     | 0.975    |
| Hybrid-FZ                | − 1.307 | − 0.837 | 0.707    |           | 0.722    | 1.670     | 1.167    |
| GM-IP-FZ                 | − 1.466 | − 1.326 | − 0.881  | − 0.722   |          | 2.255     | 0.952    |
| GM-PPI-FZ                | − 2.339 | − 2.132 | − 2.282  | − 1.670   | − 2.255  |           | − 0.473  |
| GM-RV-FZ                 | − 1.778 | − 1.665 | − 0.975  | − 1.167   | − 0.952  | 0.473     |          |
| $\alpha_{level} = 5\%$   |         |         |          |           |          |           |          |
| G2-FZ                    |         | − 0.639 | − 0.900  | − 1.143   | − 0.896  | − 0.044   | 0.703    |
| G1-FZ                    | 0.639   |         | − 0.569  | − 1.260   | − 0.562  | 0.373     | 1.164    |
| GCH-N-FZ                 | 0.900   | 0.569   |          | 0.232     | 0.133    | 2.340     | 3.027    |
| Hybrid-FZ                | 1.143   | 1.260   | − 0.232  |           | − 0.226  | 0.661     | 1.411    |
| GM-IP-FZ                 | 0.896   | 0.562   | − 0.133  | 0.226     |          | 2.353     | 3.038    |
| GM-PPI-FZ                | 0.044   | − 0.373 | − 2.340  | − 0.661   | − 2.353  |           | 1.440    |
| GM-RV-FZ                 | − 0.703 | − 1.164 | − 3.027  | − 1.411   | − 3.038  | − 1.440   |          |
| $\alpha_{level} = 10\%$  |         |         |          |           |          |           |          |
| G2-FZ                    |         | − 3.435 | − 4.885  | − 2.752   | − 4.889  | − 4.143   | − 3.551  |
| G1-FZ                    | 3.435   |         | − 3.220  | 0.772     | − 3.222  | − 2.470   | − 1.902  |
| GCH-N-FZ                 | 4.885   | 3.220   |          | 3.505     | 0.034    | 1.811     | 0.933    |
| Hybrid-FZ                | 2.752   | − 0.772 | − 3.505  |           | − 3.507  | − 2.716   | − 2.029  |
| GM-IP-FZ                 | 4.889   | 3.222   | − 0.034  | 3.507     |          | 1.814     | 0.933    |
| GM-PPI-FZ                | 4.143   | 2.470   | − 1.811  | 2.716     | − 1.814  |           | 0.197    |
| GM-RV-FZ                 | 3.551   | 1.902   | − 0.933  | 2.029     | − 0.933  | − 0.197   |          |

Notes: This table provides  $t$ -statistic of Diebold and Mariano (1995) for comparing the average losses between Patton *et al.* (2019) and our proposed models across the tailed probability level  $\alpha_{level}$  over the in sample period from January 1928 to December 2008. A positive value implies that the column model has lower average loss than the row model. The  $t$ -statistic of Diebold and Mariano (1995) is asymptotically normal and thus the critical value is easily obtained.

Table 9. Out-of-Sample (OoS) average losses and goodness-of-fit (GoF) tests.

| $\alpha_{level}$ | Panel A: average loss functions |              |              |              | Panel B: GoF test $p$ -values for VaR |              |              |              | Panel C: GoF test $p$ -values for ES |              |              |              |
|------------------|---------------------------------|--------------|--------------|--------------|---------------------------------------|--------------|--------------|--------------|--------------------------------------|--------------|--------------|--------------|
|                  | 1%                              | 2.5%         | 5%           | 10%          | 1%                                    | 2.5%         | 5%           | 10%          | 1%                                   | 2.5%         | 5%           | 10%          |
| RW125            | 1.423                           | 1.135        | 0.908        | 0.633        | 0.000                                 | 0.000        | 0.000        | 0.000        | 0.001                                | 0.000        | 0.000        | 0.000        |
| RW250            | 1.432                           | 1.152        | 0.922        | 0.658        | 0.000                                 | 0.000        | 0.000        | 0.000        | 0.000                                | 0.000        | 0.000        | 0.000        |
| RW500            | 1.480                           | 1.207        | 0.989        | 0.711        | 0.000                                 | 0.000        | 0.000        | 0.000        | 0.000                                | 0.000        | 0.000        | 0.000        |
| GCH-N-MLE        | 1.688                           | 1.175        | 0.868        | 0.571        | 0.000                                 | 0.000        | 0.000        | <b>0.313</b> | 0.000                                | 0.000        | 0.000        | 0.000        |
| GCH-SKT-MLE      | 1.385                           | 1.074        | 0.833        | 0.562        | 0.001                                 | 0.000        | 0.000        | <b>0.155</b> | 0.001                                | 0.000        | 0.000        | 0.001        |
| GCH-EDF-MLE      | 1.262                           | 0.999        | 0.788        | 0.541        | 0.006                                 | 0.012        | 0.007        | <b>0.296</b> | 0.003                                | 0.006        | 0.003        | 0.022        |
| G2-FZ            | 1.193                           | 0.967        | 0.762        | 0.516        | <b>0.132</b>                          | <b>0.236</b> | <b>0.742</b> | <b>0.716</b> | <b>0.218</b>                         | <b>0.372</b> | <b>0.270</b> | <b>0.657</b> |
| G1-FZ            | 1.254                           | 0.974        | 0.776        | 0.522        | 0.005                                 | <b>0.307</b> | <b>0.784</b> | <b>0.179</b> | 0.014                                | <b>0.153</b> | <b>0.413</b> | <b>0.497</b> |
| GCH-N-FZ         | 1.266                           | 0.996        | 0.781        | 0.533        | 0.001                                 | 0.097        | <b>0.124</b> | <b>0.365</b> | 0.001                                | 0.014        | 0.016        | 0.084        |
| Hybrid-FZ        | 1.235                           | 0.959        | 0.756        | <b>0.512</b> | 0.013                                 | 0.213        | 0.505        | 0.849        | 0.018                                | 0.150        | 0.394        | <b>0.688</b> |
| GM-IP-MLE        | 1.368                           | 1.025        | 0.779        | 0.528        | 0.001                                 | 0.038        | <b>0.815</b> | <b>0.559</b> | 0.000                                | 0.004        | <b>0.110</b> | <b>0.569</b> |
| GM-PPI-MLE       | 1.328                           | 1.006        | 0.770        | 0.524        | 0.003                                 | 0.097        | <b>0.988</b> | <b>0.378</b> | 0.001                                | 0.012        | <b>0.364</b> | <b>0.648</b> |
| GM-RV-MLE        | 1.269                           | 0.976        | 0.760        | 0.529        | 0.050                                 | <b>0.529</b> | <b>0.795</b> | 0.034        | 0.017                                | 0.123        | <b>0.548</b> | <b>0.300</b> |
| GM-IP-FZ         | 1.155                           | 0.963        | 0.763        | 0.523        | <b>0.180</b>                          | <b>0.352</b> | <b>0.883</b> | <b>0.812</b> | <b>0.136</b>                         | <b>0.206</b> | <b>0.470</b> | <b>0.611</b> |
| GM-PPI-FZ        | <b>1.140</b>                    | <b>0.946</b> | 0.755        | 0.520        | <b>0.257</b>                          | <b>0.637</b> | <b>0.982</b> | <b>0.813</b> | <b>0.171</b>                         | <b>0.396</b> | <b>0.696</b> | <b>0.663</b> |
| GM-RV-FZ         | 1.173                           | 0.954        | <b>0.754</b> | 0.527        | 0.008                                 | <b>0.231</b> | <b>0.275</b> | 0.094        | 0.008                                | 0.055        | <b>0.104</b> | 0.090        |

Notes: This table provides the average loss based on equation (3), and the  $p$  values from goodness-of-fit (GoF) tests for VaR and ES forecasts. The bold values in Panel A denote the lowest loss function at a given tail. The bold numbers in Panels B and C denote the  $p$ -values higher than 0.1. The evaluation period is from January 2009 to December 2018.



Table 10. MCS results for the evaluation period from January 2009 to December 2018.

| $\alpha_{level}$ | 1%    | 2.5%  | 5%    | 10%   |
|------------------|-------|-------|-------|-------|
| RW125            | 0.003 | 0.002 | 0.001 | 0.001 |
| RW250            | 0.038 | 0.010 | 0.001 | 0.001 |
| RW500            | 0.000 | 0.000 | 0.000 | 0.000 |
| GCH-N-MLE        | 0.000 | 0.000 | 0.000 | 0.000 |
| GCH-SKT-MLE      | 0.021 | 0.005 | 0.002 | 0.002 |
| GCH-EDF-MLE      | 0.061 | 0.113 | 0.130 | 0.025 |
| G2-FZ            | 0.296 | 0.490 | 0.684 | 0.560 |
| G1-FZ            | 0.106 | 0.322 | 0.251 | 0.324 |
| GCH-N-FZ         | 0.003 | 0.023 | 0.073 | 0.039 |
| Hybrid-FZ        | 0.059 | 0.728 | 0.920 | 1.000 |
| GM-IP-MLE        | 0.020 | 0.021 | 0.062 | 0.149 |
| GM-PPI-MLE       | 0.030 | 0.038 | 0.167 | 0.256 |
| GM-RV-MLE        | 0.060 | 0.253 | 0.725 | 0.349 |
| GM-IP-FZ         | 0.431 | 0.086 | 0.264 | 0.443 |
| GM-PPI-FZ        | 1.000 | 1.000 | 0.983 | 0.697 |
| GM-RV-FZ         | 0.217 | 0.745 | 1.000 | 0.392 |

Notes: This table shows the  $p$ -values of model confidence set test for the prediction of VaR and ES. The abbreviation of these models refer to table 4. Evaluation period is from January 2009 to December 2018.

where  $t_i = \frac{\bar{d}_i}{\sqrt{\hat{var}(\bar{d}_i)}}$ , and  $\hat{var}(\bar{d}_i)$  is the estimates of  $var(\bar{d}_i)$ .

We will obtain the critical value for the statistic (22) through the bootstrap technique, because its asymptotic distribution is claimed by Hansen *et al.* (2011) to be nonstandard.

Use a confidence level of 90%, table 10 reports the  $p$ -values of the MCS test. We find that for the 1% tail, the two-factor GAS model and the three GARCH-MIDAS with  $FZ$  minimization estimates are included in the MCS, with the  $p$ -values higher than 0.1. The remaining models under consideration are excluded, indicating that they produce VaR and ES forecasts that are on average significantly less accurate than the aforementioned four models in the MCS. For the 2.5% tail, the MCS additionally incorporates the one-factor GAS but excludes the GM-IP-FZ. For the 5% and 10% tails, the MCS includes two GAS models and three GARCH-MIDAS models with  $FZ$  estimates. Therefore, we can conclude that the GARCH-MIDAS with macroeconomic uncertainty variables and the two-factor GAS under the  $FZ$  minimization estimates significantly outperform the other competing models.

To investigate the effects of the choice of evaluation sample on forecasting performance, we consider two longer evaluation periods: the first is from January 1989 to December 2018, and the second is from January 1999 to December 2018. The MCS test results for these two periods are reported in tables 11 and 12, respectively. We find that only three volatility models are included in the MCS for all tails during each evaluation period: these models are the GARCH-MIDAS models with  $FZ$  minimization estimates (i.e. GM-RV-FZ, GM-IP-FZ and GM-PPI-FZ). In summary, our out-of-sample evidence indicates that  $FZ$  minimization estimates work better than QMLE estimates in forecasting VaR and ES. Furthermore, macroeconomic uncertainty variables are helpful for improving out-of-sample forecasting performance.

### 5.5. Robustness test and discussions

Although industrial production (IP) and producer price index (PPI) are widely used as the measures of macroeconomic

Table 11. MCS results for the evaluation period from January 1989 to December 2018.

| $\alpha_{level}$ | 1%    | 2.5%  | 5%    | 10%   |
|------------------|-------|-------|-------|-------|
| RW125            | 0.000 | 0.000 | 0.000 | 0.000 |
| RW250            | 0.009 | 0.001 | 0.001 | 0.000 |
| RW500            | 0.001 | 0.000 | 0.000 | 0.000 |
| GCH-N-MLE        | 0.001 | 0.002 | 0.005 | 0.004 |
| GCH-SKT-MLE      | 0.068 | 0.239 | 0.086 | 0.104 |
| GCH-EDF-MLE      | 0.044 | 0.254 | 0.083 | 0.106 |
| G2-FZ            | 0.087 | 0.443 | 0.537 | 0.841 |
| G1-FZ            | 0.009 | 0.016 | 0.330 | 0.823 |
| GCH-N-FZ         | 0.349 | 0.356 | 0.418 | 0.266 |
| Hybrid-FZ        | 0.031 | 0.316 | 1.000 | 1.000 |
| GM-IP-MLE        | 0.002 | 0.003 | 0.014 | 0.019 |
| GM-PPI-MLE       | 0.005 | 0.009 | 0.053 | 0.057 |
| GM-RV-MLE        | 0.004 | 0.039 | 0.273 | 0.074 |
| GM-IP-FZ         | 0.323 | 0.382 | 0.463 | 0.248 |
| GM-PPI-FZ        | 1.000 | 1.000 | 0.764 | 0.297 |
| GM-RV-FZ         | 0.269 | 0.621 | 0.561 | 0.250 |

Notes: This table shows the  $p$ -values of model confidence set test for the prediction of VaR and ES. The abbreviation of these models refer to table 4. Evaluation period is from January 1989 to December 2018.

Table 12. MCS results for the evaluation period from January 1999 to December 2018.

| $\alpha_{level}$ | 1%    | 2.5%  | 5%    | 10%   |
|------------------|-------|-------|-------|-------|
| RW125            | 0.002 | 0.007 | 0.000 | 0.000 |
| RW250            | 0.017 | 0.003 | 0.000 | 0.000 |
| RW500            | 0.001 | 0.000 | 0.000 | 0.000 |
| GCH-N-MLE        | 0.009 | 0.010 | 0.010 | 0.132 |
| GCH-SKT-MLE      | 0.057 | 0.159 | 0.311 | 0.154 |
| GCH-EDF-MLE      | 0.032 | 0.146 | 0.310 | 0.139 |
| G2-FZ            | 0.559 | 0.235 | 0.587 | 0.328 |
| G1-FZ            | 0.283 | 0.528 | 0.642 | 1.000 |
| GCH-N-FZ         | 0.235 | 0.169 | 0.432 | 0.182 |
| Hybrid-FZ        | 0.060 | 0.469 | 0.801 | 0.740 |
| GM-IP-MLE        | 0.005 | 0.002 | 0.023 | 0.122 |
| GM-PPI-MLE       | 0.039 | 0.023 | 0.129 | 0.314 |
| GM-RV-MLE        | 0.075 | 0.100 | 0.271 | 0.226 |
| GM-IP-FZ         | 0.277 | 0.167 | 0.383 | 0.156 |
| GM-PPI-FZ        | 1.000 | 1.000 | 1.000 | 0.244 |
| GM-RV-FZ         | 0.044 | 0.387 | 0.578 | 0.274 |

Notes: This table shows the  $p$ -values of model confidence set test for the prediction of VaR and ES. The abbreviation of these models refer to table 4. Evaluation period is from January 1999 to December 2018.

fundamentals, they do not fully reflect the economic activities. Alternatively, we use a comprehensive policy-related economic policy uncertainty (EPU) proposed by Baker *et al.* (2016). The data can be collected from the website [†](https://www.policyuncertainty.com/index.html). To match with the data availability for EPU index, our sample period covers from December 1, 1984 to December 1, 2018, yielding 408 monthly observations. The first 29 years observations are for estimation, and the remaining 5 years are for out-of-sample analysis.

Table 13 provides the out-of-sample average losses and goodness-of-fit tests for the new data. We observe that the GM-EPU-FZ model performs best for extreme situation, with

[† https://www.policyuncertainty.com/index.html](https://www.policyuncertainty.com/index.html).



Table 13. Out-of-Sample (OoS) average losses and goodness-of-fit (GoF) tests.

| $\alpha_{level}$ | Panel A: average loss functions |              |              |              | Panel B: GoF test $p$ -values for VaR |              |              |              | Panel C: GoF test $p$ -values for ES |              |              |              |
|------------------|---------------------------------|--------------|--------------|--------------|---------------------------------------|--------------|--------------|--------------|--------------------------------------|--------------|--------------|--------------|
|                  | 1%                              | 2.5%         | 5%           | 10%          | 1%                                    | 2.5%         | 5%           | 10%          | 1%                                   | 2.5%         | 5%           | 10%          |
| RW125            | 1.395                           | 1.044        | 0.806        | 0.497        | 0.000                                 | 0.021        | 0.002        | 0.054        | 0.002                                | 0.004        | 0.011        | 0.015        |
| RW250            | 1.241                           | 0.986        | 0.757        | 0.484        | 0.015                                 | <b>0.265</b> | 0.072        | 0.005        | 0.021                                | <b>0.168</b> | 0.027        | 0.002        |
| RW500            | 1.297                           | 1.058        | 0.852        | 0.568        | 0.061                                 | 0.027        | 0.006        | 0.000        | <b>0.130</b>                         | 0.055        | 0.009        | 0.000        |
| GCH-N-MLE        | 1.295                           | 0.898        | 0.626        | 0.368        | 0.069                                 | <b>0.621</b> | <b>0.315</b> | 0.008        | 0.017                                | <b>0.143</b> | <b>0.171</b> | 0.073        |
| GCH-SKT-MLE      | 1.119                           | 0.854        | <b>0.613</b> | 0.362        | 0.087                                 | <b>0.560</b> | <b>0.371</b> | <b>0.294</b> | 0.010                                | 0.089        | 0.065        | <b>0.190</b> |
| GCH-EDF-MLE      | 1.120                           | 0.850        | <b>0.613</b> | 0.362        | <b>0.106</b>                          | <b>0.385</b> | <b>0.315</b> | <b>0.221</b> | 0.013                                | 0.034        | 0.048        | <b>0.119</b> |
| G2-FZ            | 1.090                           | 0.878        | 0.636        | <b>0.354</b> | <b>0.331</b>                          | <b>0.462</b> | <b>0.263</b> | 0.088        | <b>0.193</b>                         | 0.015        | 0.037        | 0.046        |
| G1-FZ            | 1.121                           | 0.902        | 0.650        | 0.374        | 0.078                                 | <b>0.170</b> | <b>0.116</b> | 0.083        | 0.015                                | 0.010        | 0.012        | 0.054        |
| GCH-N-FZ         | 1.080                           | 0.840        | 0.616        | 0.368        | 0.073                                 | <b>0.409</b> | <b>0.356</b> | <b>0.159</b> | 0.010                                | 0.049        | 0.070        | <b>0.117</b> |
| Hybrid-FZ        | 1.139                           | 0.890        | 0.635        | 0.366        | <b>0.239</b>                          | <b>0.295</b> | <b>0.232</b> | <b>0.198</b> | <b>0.102</b>                         | 0.016        | 0.033        | 0.097        |
| GM-IP-MLE        | 1.286                           | 0.893        | 0.622        | 0.366        | 0.072                                 | <b>0.595</b> | <b>0.420</b> | 0.008        | 0.019                                | <b>0.144</b> | <b>0.249</b> | 0.068        |
| GM-PPI-MLE       | 1.286                           | 0.893        | 0.623        | 0.366        | 0.073                                 | <b>0.594</b> | <b>0.420</b> | 0.008        | 0.019                                | <b>0.143</b> | <b>0.248</b> | 0.068        |
| GM-RV-MLE        | 1.455                           | 0.950        | 0.645        | 0.373        | 0.022                                 | <b>0.412</b> | <b>0.561</b> | <b>0.364</b> | 0.004                                | 0.059        | 0.083        | <b>0.171</b> |
| GM-EPU-MLE       | 1.286                           | 0.893        | 0.622        | 0.366        | 0.072                                 | <b>0.595</b> | <b>0.420</b> | 0.008        | 0.019                                | <b>0.144</b> | <b>0.249</b> | 0.068        |
| GM-IP-FZ         | 1.077                           | 0.841        | 0.618        | 0.371        | 0.060                                 | <b>0.459</b> | <b>0.505</b> | <b>0.183</b> | 0.008                                | 0.057        | <b>0.242</b> | <b>0.100</b> |
| GM-PPI-FZ        | 1.080                           | 0.845        | 0.618        | 0.369        | 0.074                                 | <b>0.393</b> | <b>0.386</b> | <b>0.192</b> | 0.011                                | 0.036        | <b>0.297</b> | <b>0.103</b> |
| GM-RV-FZ         | 1.198                           | 0.876        | 0.624        | 0.366        | 0.023                                 | <b>0.282</b> | <b>0.803</b> | <b>0.538</b> | 0.002                                | 0.009        | <b>0.183</b> | <b>0.239</b> |
| GM-EPU-FZ        | <b>1.059</b>                    | <b>0.834</b> | 0.616        | 0.368        | 0.069                                 | <b>0.393</b> | <b>0.284</b> | 0.062        | 0.011                                | 0.057        | 0.065        | 0.055        |

Notes: This table provides the average loss based on equation (3), and the  $p$  values from goodness-of-fit (GoF) tests for VaR and ES forecasts. The bold values in Panel A denote the lowest loss function at a given tail. The bold numbers in Panels B and C denote the  $p$ -values higher than 0.1. The evaluation period is from January 2014 to December 2018.

Table 14. Estimation results based on robust regression.

| dependent variable | IP volatility |            | PPI volatility |            | Realized Volatility (RV) |            |
|--------------------|---------------|------------|----------------|------------|--------------------------|------------|
|                    | estimate      | s.e.       | estimate       | s.e.       | estimate                 | s.e.       |
| $\alpha_h$         | 5.14E-05***   | (3.03E-06) | 2.57E-05***    | (1.47E-06) | 1.42E-03***              | (4.68E-05) |
| $\beta_h$          | 3.36E-05***   | (7.08E-06) | 1.58E-05***    | (3.43E-06) | 1.53E-03***              | (1.09E-04) |

Notes: This table provides the results for the Huber (1964) robust regression over the period is from January 1928 to December 2018. The asterisks \*, \*\*, and \*\*\* indicate significance at the 10%, 5%, and 1% levels, respectively.

the lowest average loss values of 1.059 for 1% quantile and 0.834 for 2.5% quantile. This is in line with the conclusion we provide above, and the EPU index as the uncertainty proxy is more powerful than the traditional macroeconomics index, such as industrial production (IP) and producer price index (PPI).

We further investigate why the models with macroeconomic uncertainty outperform the competing models in extreme circumstances. We interpret that the magnitude of stock volatility is extremely strengthened by macroeconomic uncertainty during the period of recession, which leads to more accurate measure for VaR and ES. To exhibit this, we regress macroeconomic uncertainty variable ( $y_t$ ) on the NBER dated business cycle indicator. The data are collected from Federal Reserve Bank at Saint Louis<sup>†</sup>, with the sample period from January 1, 1928 through December 1, 2018.

$$y_t = \alpha_h + \beta_h 1_{rec,t} + \epsilon_t \quad (23)$$

where  $1_{rec,t}$  is an indicator function which takes the value of unity during a recession period, and zero otherwise. If the coefficient on  $\beta_h$  is positive and significant, we then can conclude that the macroeconomic uncertainty is greater for

recessionary period than normal period. Notice that there exist many outliers for the macroeconomic uncertainty variables shown in figure 1. We use robust regression of Huber (1964) with the objective function as

$$\rho_h(\hat{\epsilon}) = \begin{cases} \frac{1}{2}\hat{\epsilon}^2, & \text{for } |\hat{\epsilon}| \leq M \\ k|\hat{\epsilon}| - \frac{1}{2}k^2, & \text{for } |\hat{\epsilon}| > M \end{cases} \quad (24)$$

and the weighting function as

$$\omega_h(\hat{\epsilon}) = \begin{cases} 1, & \text{for } |\hat{\epsilon}| \leq M \\ \frac{1}{|\hat{\epsilon}|}, & \text{for } |\hat{\epsilon}| > M \end{cases} \quad (25)$$

where  $\hat{\epsilon}$  is the residual.  $M$  is called tuning constant and is set to 1.345 using the rule of thumb. Huber regression is robust to outliers through giving different penalty for small and large residuals, see Huber (1964) for more details.

Table 14 presents the results for the huber estimates. As expected, the coefficients on  $\beta_h$  are positive and significant at level 1% for all three macroeconomic uncertainty variables, which implies that the volatility has greater magnitude during recessionary period and is transited to the total variance by our specification in (8).

<sup>†</sup> <https://fred.stlouisfed.org/>.

## 6. Conclusion

Forecasting VaR and ES has important implications for risk management. Patton *et al.* (2019) propose a dynamic modeling framework based on a Fissler and Ziegel (2016) loss function that makes VaR and ES jointly elicitable. In the framework of PZC (2019), we propose a class of GARCH-MIDAS models for VaR and ES, the parameters of which are obtained by minimizing the *FZ* loss function. We show the asymptotic theory of the new model. The Monte Carlo simulation results suggest that the estimates based on the *FZ* minimization method are more robust to model misspecification than QMLE estimates.

The empirical results are also of value. Using the proposed model, we find that macroeconomic uncertainty plays an important role in affecting the long-term volatility component. Our out-of-sample evidence indicates that accounting for the effects of macroeconomic uncertainty can significantly improve the forecasting accuracy of VaR and ES. The proposed GARCH-MIDAS based on *FZ* minimization parameters performs better than more than a dozen competing models, including GARCH, GAS, and rolling window methods.

## Disclosure statement

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## 7. Supplemental data

Supplemental data for this article can be accessed at <http://dx.doi.org/10.1080/14697688.2020.1862418>.

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